

Eilaf Alharbi
Henaf Aldughairi
Shatha Alzughibi
Lama Almushyqih



College of Computer, Qassim University
Supervisor: Dr. Hedia ZARDI

Title

Automated Inventory and Warehouse Tracking System

Introduction

Traditional inventory systems rely heavily on manual counting and periodic updates, leading to human errors, stock discrepancies, and delayed decision-making. JAREED addresses these challenges by providing real-time inventory visibility through simulated IoT sensing and automated data processing.

Objective

- Automate inventory quantity calculation
- Detect low-stock conditions
- Maintain a Digital Twin of warehouse inventory
- Forecast demand for the next 7 days
- Provide a real-time monitoring dashboard

Methodology

Approach

Prototype-based system development using 2TUP methodology

Technologies

Flask REST API, Supabase, JavaScript, Linear Regression

Sensor Inputs

Simulated RFID and Load Cell readings

Architecture

IoT Simulation → Backend Processing → Digital Twin Database → Dashboard → AI Forecasting

Key Variable

Backend

Processes sensor data and triggers alerts

Database

Stores real-time and historical inventory data

Digital Twin

Continuously reflects the current warehouse state

AI Module

Generates 7-day demand forecasts using Linear Regression

Result

Inventory Tracking Efficiency

Database

Synchronization

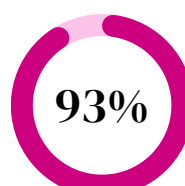
91%

Test Case Success

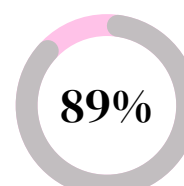
Rate

87%

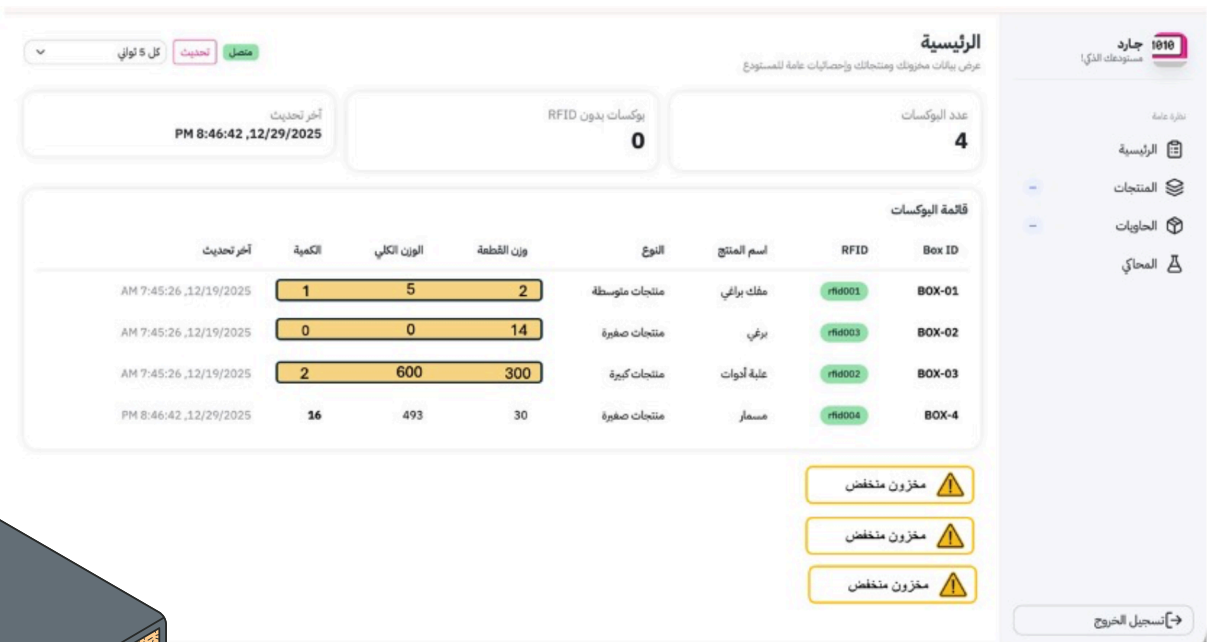
System Performance



Real-Time Updates



Low-Stock Alert



Conclusion:

JAREED successfully validates a prototype smart inventory system that combines simulated IoT sensing, Digital Twin technology, and AI-based forecasting to deliver real-time monitoring, automated alerts, and predictive inventory insights.

